Moustafa Eid

Mechanical Design Engineer

Portfolio – Mechanical Design

About Me:

I'm an Egyptian Mechanical Engineer who is passionate about Engineering Design, a primary interest in product development and machine design. I have completed my Bachelor's degree in Mechanical Engineering from Tanta University in June 2013. I believe in continuous improvement so I'm always learning on the job, and spend my free time on side projects and courses that earn me new skills. The upcoming projects provide an overview of various work experiences, and related activities that I have done since the start of my career path.



Here is a shortlist of valuable skills that contributed to fruition of the projects that I worked on:

Mechanical Design: Solidworks Expert, Creo Parametric.

Simulation: Solidworks Simulation, ANSYS mechanical.

Manufacturing: SolidCAM
Subtractive: CNC Milling, Lathe, Laser Cutting, Waterjet Cutting, Grinding, Drilling, Boring, Reaming, EDM.
Additive: 3D printing; SLA, DLP, FDM, Polyjet.

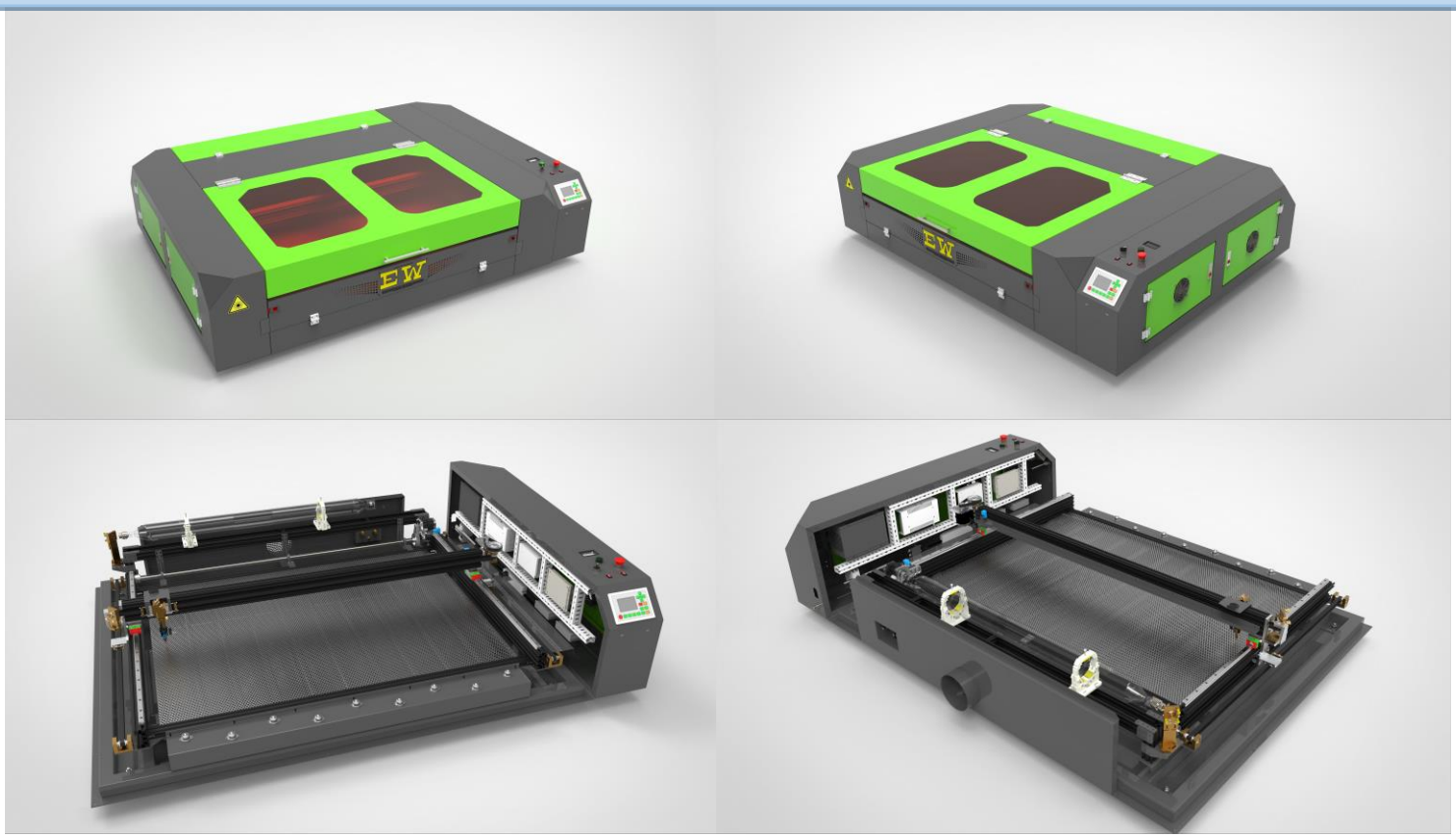
Shop Tools: MIG, TIG & Arc Welding machines, Drill Press, Sanders, Disc Cutters, Impacts & Drills, Hand Tools.

Programming: Arduino, Python.

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CO2 Laser Cutter



Project:

- Design & Fabrication of an industrial Co2 Laser Cutter.
- Laser Source Power: up to **130 Watts**.
- Acceleration Profile: up to **2000mm/s² on Hybrid Motor**.
- Power Transfer System: **Belts & Pulleys**.
- Water Chiller Cooled Tube.
- Working Area: **1300x900**mm
- **Honeycomb** Cutting Table.
- **Handling Rollers**: for thick sheets.
- Operation Voltage: **220V**.
- **Safety** Switches: Emergency, Circuit Breakers, Amp meter, On/Off, Laser Switch, Light Switch.
- Eye Protection **Door Shields**.
- **Air Assisted** Cutting.
- Air Cooled Electronics Compartment.
- **Fuse-Protected** Power Input.
- Hard & Soft **Limits**.

Process:

- Used **Solidworks Weldments** and **SheetMetal** mainly to design the machine.
- Performed **needs analysis** and Sourced different parts e.g., "Laser Tube, Mirrors Mounts, Control System and Mechanical System parts".
- Performed **FEA & CFD** analysis for design validation.
- Implemented **DFM** & **DFA** principles to reduce assembly and manufacturing cost.
- Prepared Drawings for Manufacturing and applied **GD&T** on CNC machined Parts.
- Contacted Manufacturers to quote parts for manufacturing.
- Used **TIG** & **MAG** welding to join Machine Parts.

Result:

- Reduced manufacturing cost by **30%** due to elimination of machine's lower compartment that used around **150Kg** of steel sheets.
- Reduced manufacturing and welding hours.
- Increased machine **stability** due to increased rigidity and low **Center of Mass**.
- Machine can be installed in **tight entrance spaces** due to its low height and modular design.

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Foil Rewinding Machine



Click here to watch the machine in action

Project:

- Design & Fabrication of a Foil Rewinding Machine.
- Main Rewinding Motor Power: 2 **hp**.
- Motor Control: VFD Inverter.
- Output RPM: up to **350 RPM**.
- Power Transfer System: V **Belts & Pulleys**.
- Rewind Materials: Aluminum Foil, Stretch Wrap and Baking Paper.
- Max Rewinding Width: **600**mm.
- Max Input Foil Roll Weight: **200 kg**.
- **Double Core Handling Head**.
- **Electronic Controlled winding length**.
- Operation Voltage: **220V**.
- **Circuit Breaker-Protected** Power Input.

Process:

- Used **Solidworks Weldments** and **SheetMetal** mainly to design the machine.
- Performed **needs analysis** and Sourced different machine parts.
- Designed different **machine mechanisms**.
- Performed **FEA & Frequency** analysis for design validation.
- Implemented **DFM** & **DFA** principles to reduce assembly and manufacturing cost.
- Prepared Drawings for Manufacturing.
- Used **SolidCAM** for toolpath generation for Machining.
- Used **MAG** & ARC welding to join Machine Parts.

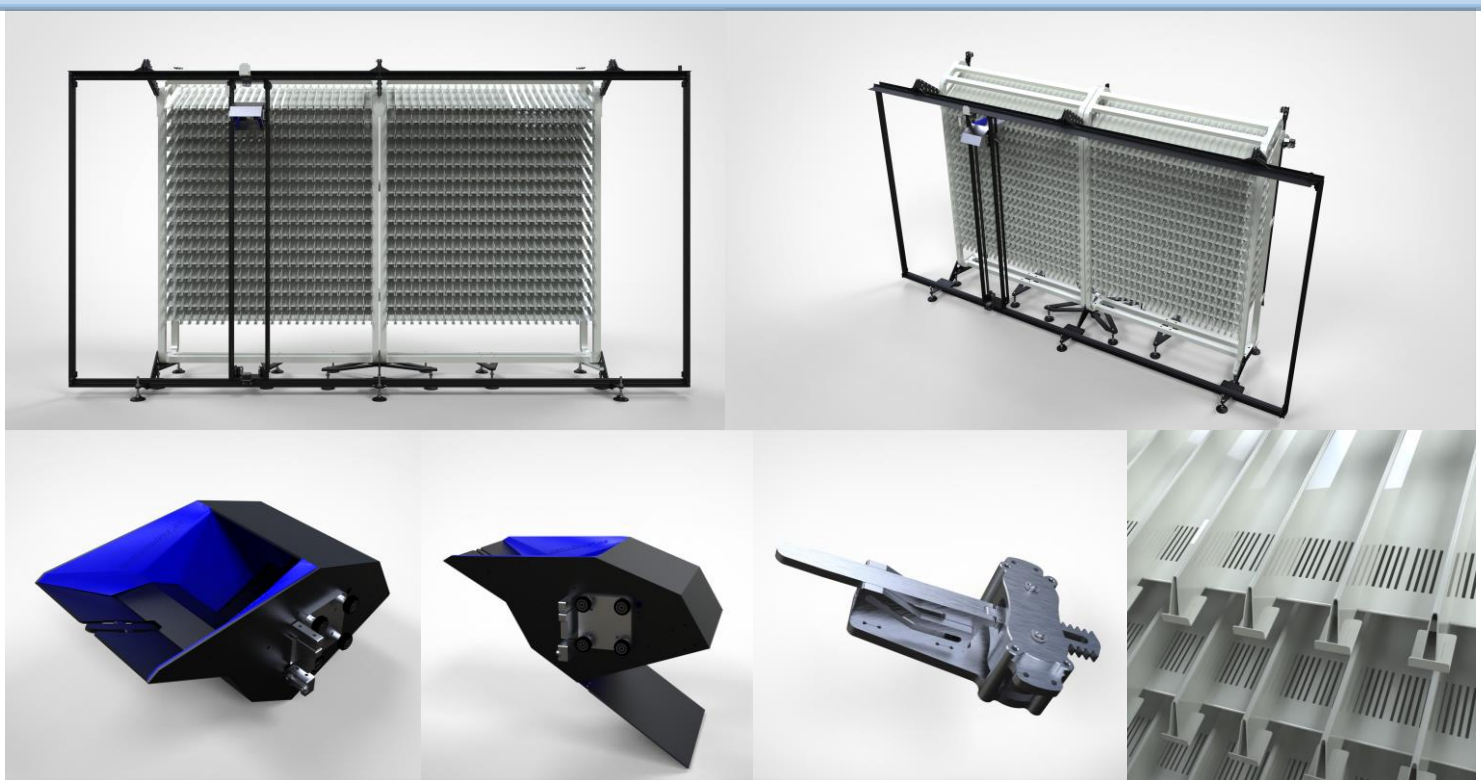
Result:

- The Machine fulfilled customer's needs rewinding up to 500kg per day.
- The Machine is designed to be converted to **Semi-Automatic** easily which increases machine efficiency and reliability.
- The Machine included **375 Liters** internal storage which lower's space required to store rewinding items.

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Electronic Pharmacy



Click here to watch a Video for the grabber mechanism in action.

Project:

- Design & Fabrication of a Concept Electronic Pharmacy Robot Prototype.
- Module Capacity over **800 medicines**.
- Grabber Robot Speed: up to **450mm/s**.
- Power Transfer System: **Belts & Pulleys**.
- Grabber Working Area: **4300x1385**mm.
- Medicine Refill by means of **Gravity**.
- **Modular Design** can be assembled onsite and customized according to area.
- Customizable medicine Slot width.
- Operation Voltage: **220V**.
- **Fuse-Protected** Power Input.
- Hard & Soft **Limits**.

Process:

- Used **Solidworks Weldments** and **SheetMetal** mainly to design the machine.
- Contributed in the ideation process of different Robot Mechanisms.
- Performed **needs analysis** and Sourced different parts e.g., "V-Slot Rails, Closed-Loop Hybrid Motors, Mounts, and Mechanical System parts".
- Performed **FEA** analysis for design validation.
- Performed **Frequency Analysis** for possible mode shapes.
- Implemented **DFM** & **DFA** principles to reduce assembly and manufacturing cost.
- Prepared Drawings for Manufacturing.
- Contacted Manufacturers to quote parts for manufacturing.
- Used **TIG** & **MAG** welding to join Robot Parts.

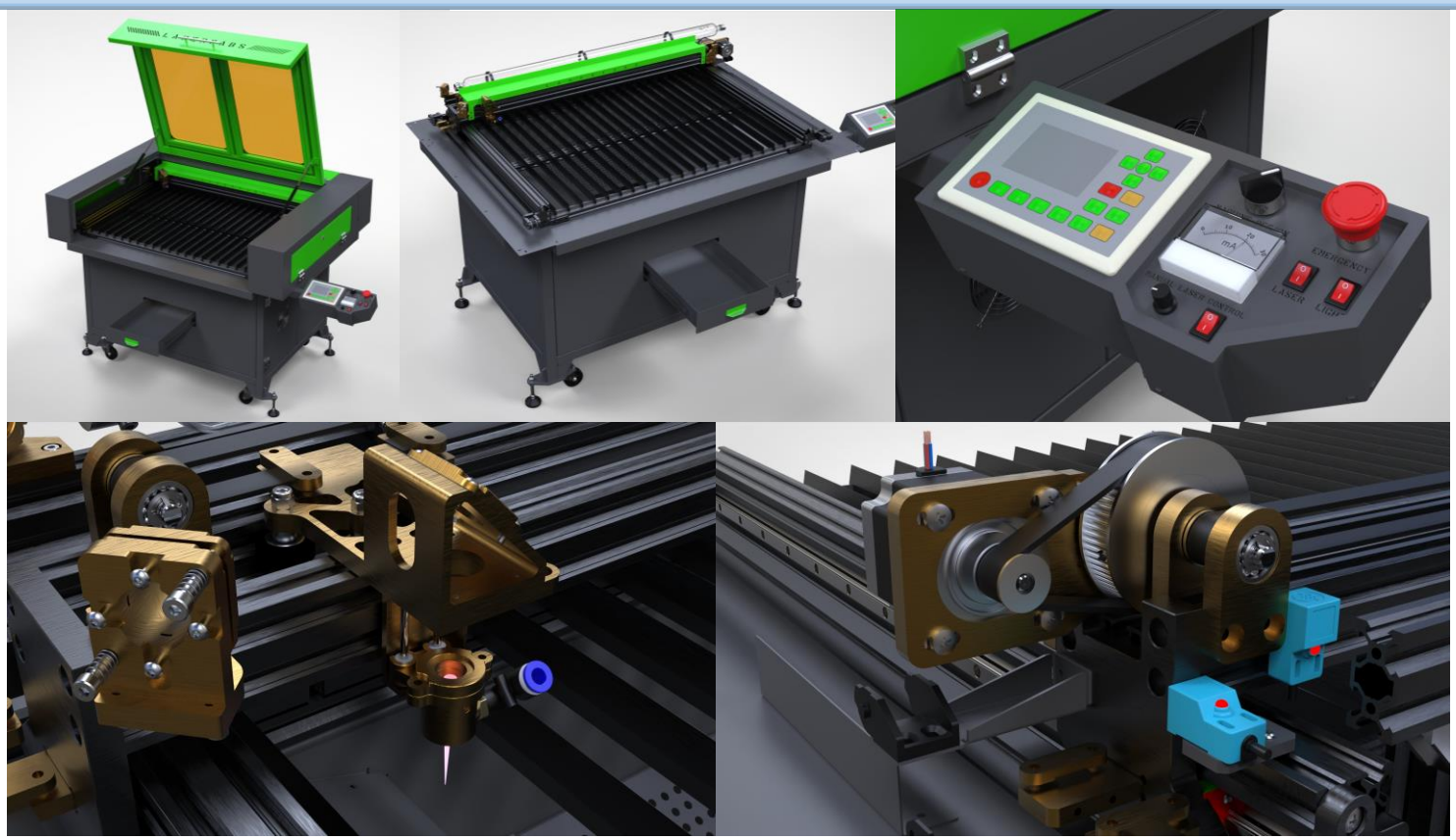
Result:

- The Prototype fulfilled its purpose with great output accuracy.
- Prototype assembly was very smooth due to its **modularity**.
- Reduced manufacturing cost due to **parts repeatability**.
- Thanks to **precise design approach**, manufacturing defects and flaws were greatly reduced which resulted in a **highly accurate steel fabrication** that can be assembled using **bolts and screws** with minimal errors.
- The Robot can be **easily transported** as it can be disassembled into easy-to-handle pieces and reassembled with **very low** assembly repeatability error.

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CO2 Laser Cutter Concept



Project:

- Concept Design of an industric Co2 Laser Cutter.
- Laser Source Power: up to **80 Watts**.
- Acceleration Profile: up to **3000mm/s² on Hybrid Motor**.
- Power Transfer System: **Belts & Pulleys**.
- Water Chiller Cooled Tube.
- Working Area: **900x600**mm.
- **Knife Blade** Cutting Table.
- Operation Voltage: **220V**.
- **Safety** Switches: Emergency, Circuit Breakers, Amp meter, On/Off, Laser Switch, Light Switch.
- **Manual Laser Power Control.**
- Eye Protection **Door Shields**.
- **Air Assisted** Cutting.
- Air Cooled Electronics Compartment.
- **Fuse-Protected** Power Input.
- Hard & Soft **Limits**.

Process:

- Used **Solidworks Weldments** and **SheetMetal** mainly to design the machine.
- Performed **needs analysis** and Sourced different parts e.g., "Laser Tube, Mirrors Mounts, Control System and Mechanical System parts".
- Performed **FEA & CFD** analysis for design validation.
- Implemented **DFM** & **DFA** principles to reduce assembly and manufacturing cost.

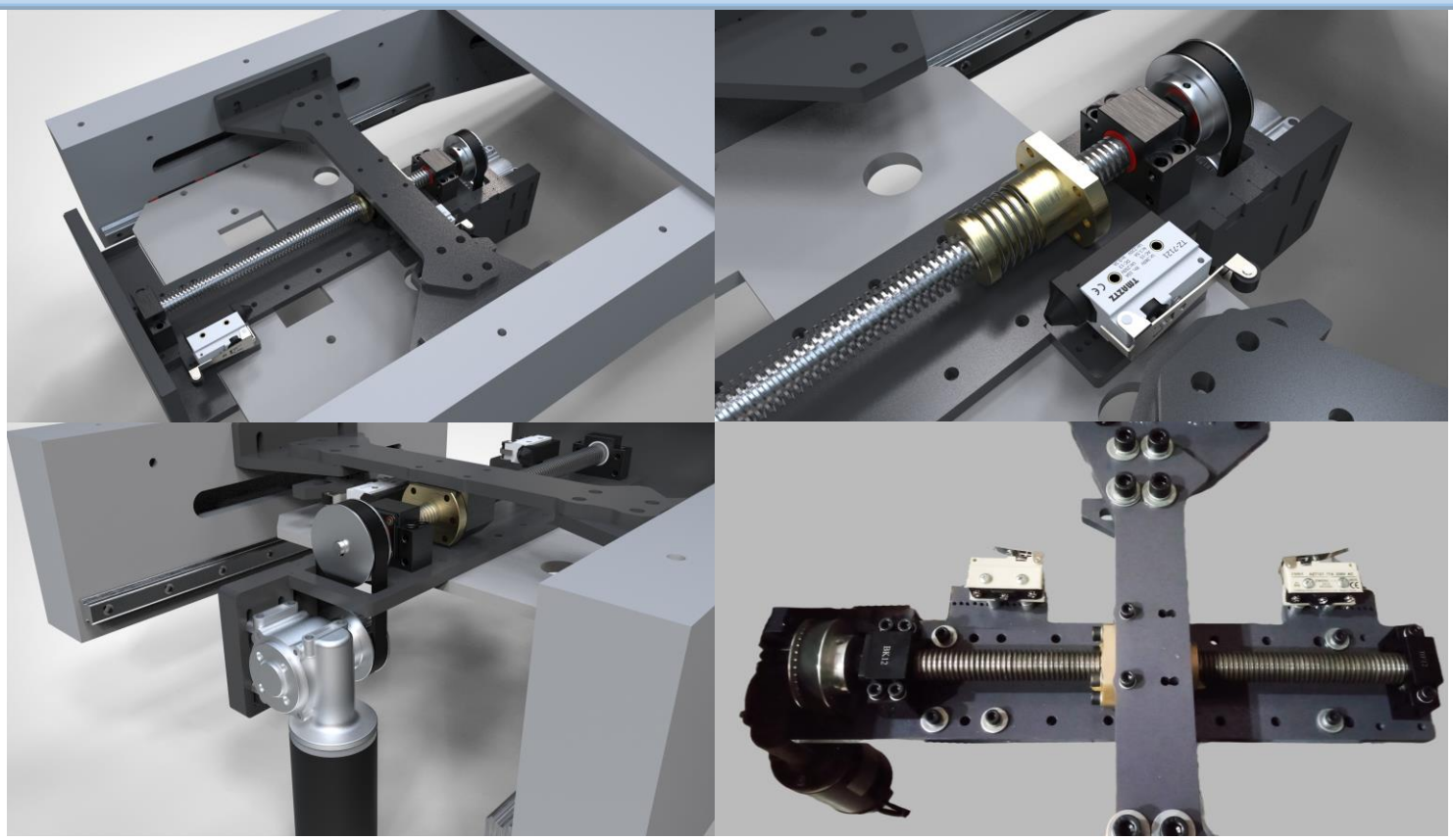
Result:

- Designed a new concept, **low weight** optic mounting system to **reduce inertia which increases motors efficiency**.
- Designed a new concept **focal length adjust** that can be automated.
- Introduce Aluminum V-Slot **C-Beam** gantry for X & Y Axis for increased rigidity while maintaining **low weight**.
- Machine maintenance is facilitated by adding side and back doors.
- Added Exhaust Fan Input to eliminate cutting fumes.
- Added quick air connector and water chiller hose input/output for easy handling of machine items.

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Medical Operations Table Upgrade



Project:

- Design an **upgrade** for a sliding mechanism of a **medical operations table**.
- Old Sliding mechanism handles a **max** weight of **250Kg**, the new Sliding mechanism shall handle over **400Kg**.
- Extend Sliding Stroke from **200mm** to **350mm** without affecting **center of mass**.
- Power Transfer System: **Lead/Ball Screw,** rather than old **Rack & Gear linear motor Mechanism.**
- **Anti-Backdrive mechanism.**
- Mechanism should be able to withstand up to **5000N of thrust force.**
- Operation Voltage: **24V DC**.
- Hard **Limits**.

Process:

- Used **Solidworks SheetMetal** mainly to design the upgrade.
- Performed **needs analysis** and Sourced different parts e.g., "Ball Screw, Screw Fixture Mounts, Limit Switch and WG Motor".
- Performed **FEA analysis** and **Bolt analysis** for design validation.
- **Selected** Lead/Ball Screws and calculated their efficiency and **required torque** based on **allowable axial load**, **thrust force required** and **screw lead**.
- Prepared Drawings for Fabrication.
- Used **MAG** welding to join Steel Parts.

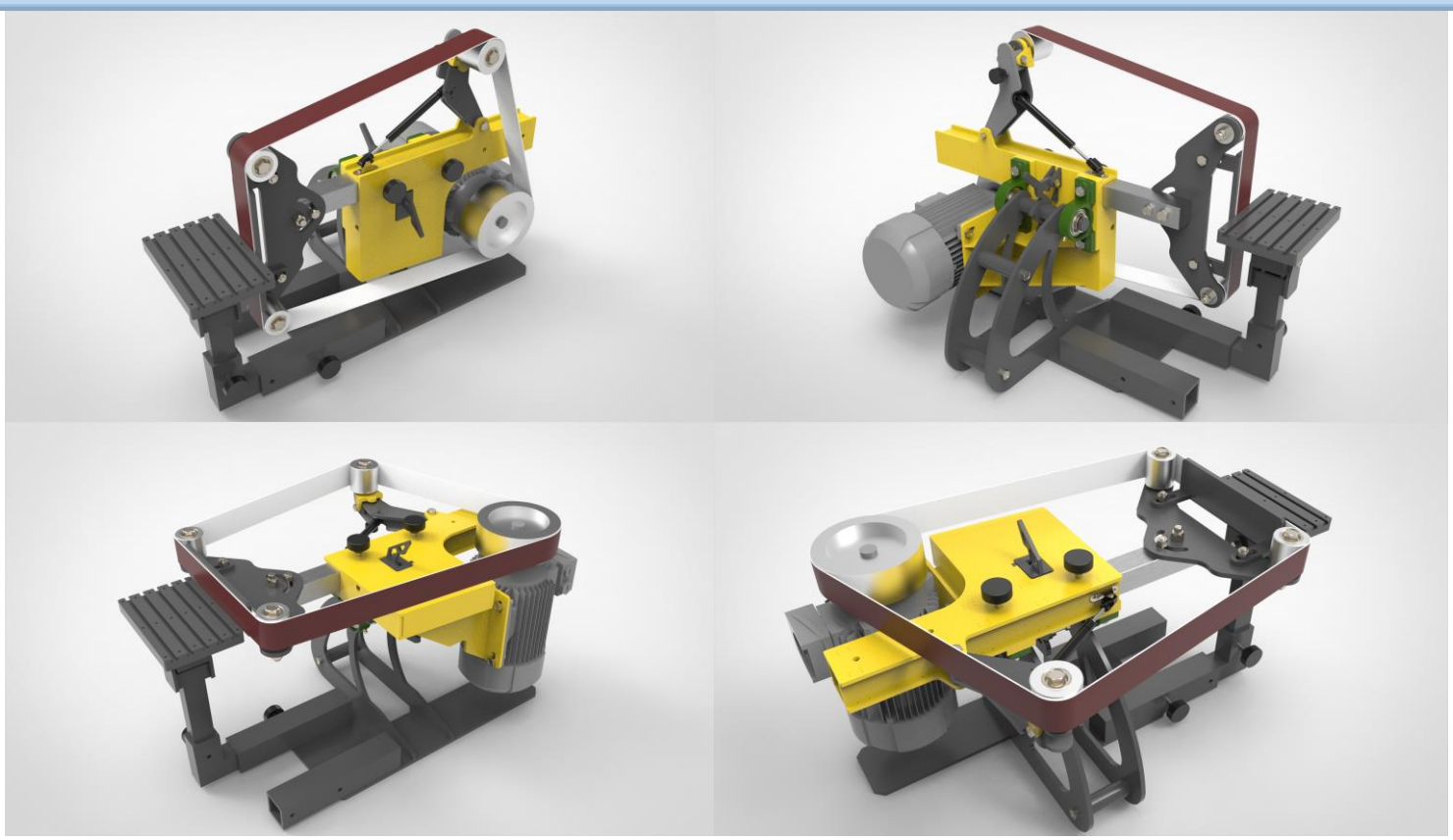
Result:

- Designed 2 mechanisms, first one is a lead screw with a lead less than **1/3** of the diameter to achieve **self-anti-backdrive**.
- The second mechanism utilized a **ball screw** which increased efficiency significantly, therefore the mechanism is likely to Backdrive, so introduced a **worm gear** irreversible mechanism to prevent backdrive.
- Utilized a **Fixed-Support** method for screw support.
- Utilized **belt-pulley** power transfer unit **between motor and screw** as a safety factor in case of hard limits failure.

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Concept Belt Grinder



Project:

- Design a 2x72'' Belt grinder concept.
- Horizontal Tilting feature.
- VFD Control.
- T-Slot Tilting Work Rest.
- Operation Voltage: 220V.

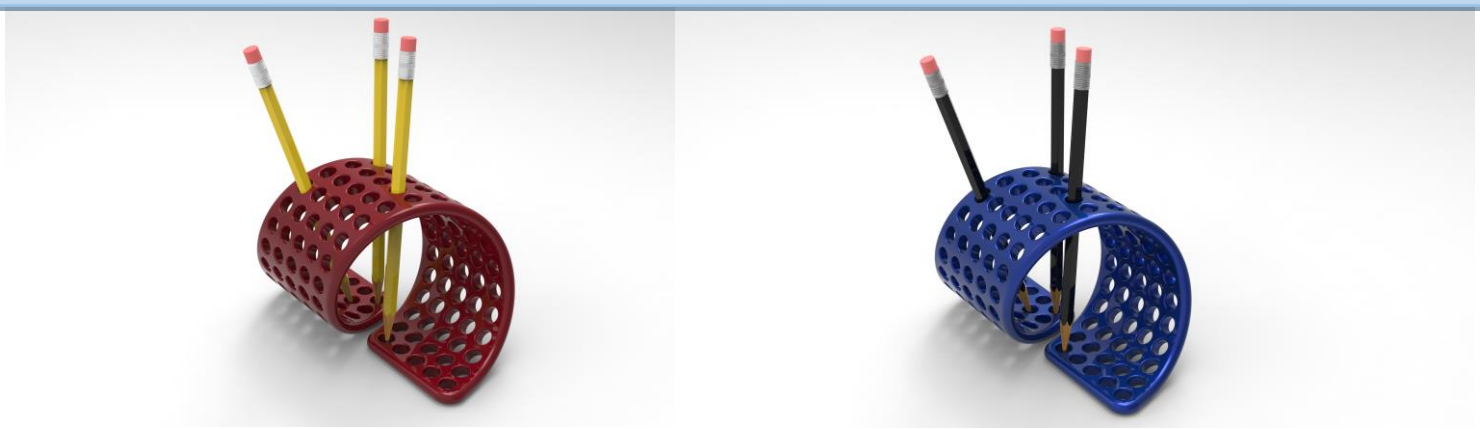
Process:

- Used **Solidworks SheetMetal** mainly to design the upgrade.
- Performed **needs analysis** and Sourced different parts.
- Performed **FEA analysis** for design validation.
- Performed **Frequency analysis** for possible mode shapes.

Result:

- Designed an **easy to assemble**, yet **rigid and sturdy** belt grinder.
- Designed a **position locking mechanism** for horizontal and vertical positions locking.
- Utilized minimal sheetmetal **thickness variation** to minimize fabrication cost.
- Added **tilting platen** feature.

Pencil Case Design



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Functional Wooden Locomotive Toy



Click here to watch a Simulated video for the locomotive mechanism

Project:

- Design a functional wooden locomotive toy.
- **Wind-up motor** driven; **Rubber Band**.

Process:

- Used **Solidworks** to design the locomotive.
- Made **internet research** to extract all mechanisms of a locomotive.
- Created a **list of requirements** to start a task-based design.
- **Sketched** different mechanisms **ideas** and selected the most suitable ones.
- Utilized **Tap and Slot** feature to facilitate parts assembly.
- Optimized parts for **Laser Cutting**.
- Created **cutout sheets**.

Result:

- Utilized **gears** as main drive power transfer mechanism.
- Designed a **locking mechanism** for the main drive gear and added a **toggle mechanism** on locking for a better user experience.
- Designed a **Cam-Follower** mechanism to simulate train's smoke stack.
- Designed a train-carriage **coupler mechanism**.
- Designed an **open-close mechanism** for the carriage door using gears and rack.
- Total parts count: **450 Parts**

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Surfacing Projects













Project:

- Complex Surfacing Projects.
- High Quality Surface output.

Process:

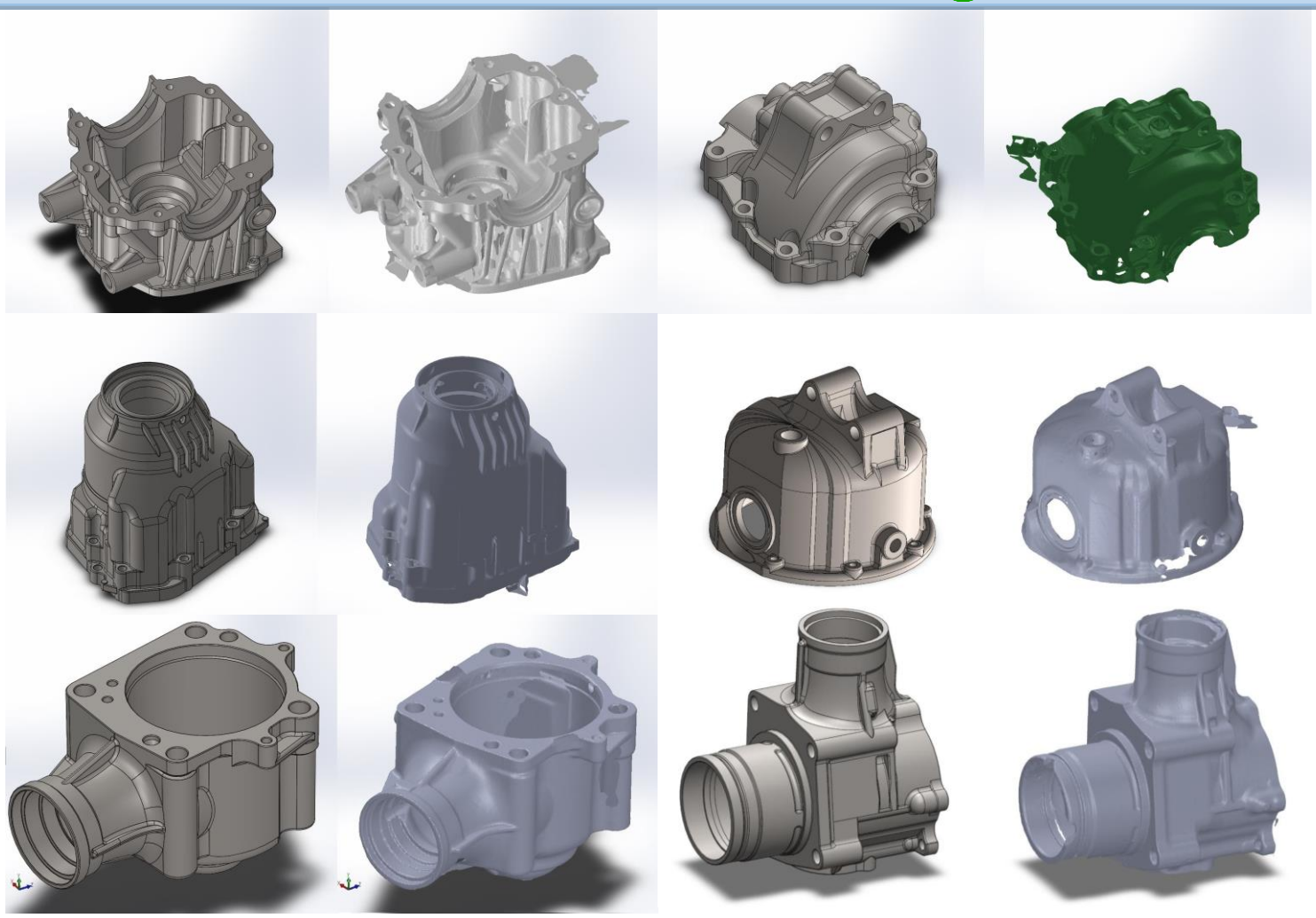
- Used **Solidworks Surfacing** mainly to design these projects.
- Performed **Surface Quality Checks** using **Visual Inspection, Zebra Stripes, Deviation Analysis, Curvature Combs, Model Curvature** and **Tangent Edge Visibility**.
- Converted surfaces to solids to create **shell bodies**.
- Final surface quality recheck.



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3D Scan Mesh Parts Conversion to Geometric Parts – Upwork



Project:

- Convert **3D Scanned** parts to industrial geometric parts.
- Parts are provided in **X_T** and **STL** formats.

Process:

- Used **Solidworks** to read X_T and STL files.
- Noticed that X_T files provide some **reference data** that could be very useful to start building up the model.
- Used STL files as reference for **extracting dimensions**.
- Built up 3d models while monitoring, comparing and judging scan files with a mechanical engineer's eye

Result:

- The output designs brought client's immense happiness and trust.
- Designs were delivered on-time.
- New projects were awarded from the same client.